MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH PROCTORED ONLINE MAKEUP EXAMINATIONS, MARCH 2022

INTRODUCTION TO LIGHTING DESIGN [ELE 4302]

REVISED CREDIT SYSTEM

Time: 75 Minutes + 10 Minutes **Date: 01 March 2022** **Max. Marks: 20**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Time: 75 minutes for writing + 10 minutes for uploading.

1A. Two lamps A and B are 6m apart. Lamps A and B are mounted at 3m and 5m above the ground respectively. Both lamps uniformly emit light in all directions. Lamp A emits 300 candelas and Lamp B emits 500 candelas.

Calculate:

1. Illumination directly under each lamp on the ground level.
2. Illumination at a point 2m away from the base of Lamp A on the ground level. ***(04)***

1B. With the help of necessary equations and sketches prove that "The frequency of the emitted light by LEDs depends on the band-gap energy of their p-n junction". ***(03)***

1C. Explain the process of designing a "Daylight Harvesting System". ***(03)***

2A. With a help of neat sketch explain the working principle of an equipment that measures total luminous flux instantly. ***(03)***

2B. A hall 32 m long and 16 m wide with a ceiling height of 5 meters is to be provided with a general illumination of 150 lumens/m^2. Taking a coefficient of utilization 0.56 and a maintenance factor of 0.70, determine the number of LED tube lights required, their spacing, mounting height and total wattage. Take luminous efficiency of fluorescent tube as 50 lumens/watt for a 50 W LED tube. Also, draw the lighting layout with lamp spacing. Assume a work plane height of 0.85m. ***(04)***

2C. With the help of a suitable example, explain the procedure to assess the performance of a lighting system. ***(03)***